

Dual Monte Carlo uncertainty framework for neural networks: application to a quantum voltage standard

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INTRODUCTION

Programmable Josephson Voltage Standards (PJVS) [1] enable traceable DC voltage realization with extremely low uncertainty. Neural-network-based digital twins can model PJVS behavior for prediction and optimization, but their uncertainty contributions must be quantified in compliance with the GUM [2] for metrological use. This work proposes the Dual Monte Carlo Uncertainty Framework (DMCUF), which combines GUM-compliant Monte Carlo [3] propagation of measurement uncertainties with Monte Carlo Dropout [4]—to estimate model uncertainty—and bias uncertainty evaluation from reference measurements. The method is demonstrated using DC voltage measurements from a PJVS, part of a neural-network-based digital twin developed within the scope of the EURAMET ViDiT project [5] to optimize the system and reduce setup and measurement times.

METHOD

The datasets were generated using measurements of a DC voltage step of 1.000108668 V with a standard deviation $\sigma_{\text{input}} = 79$ nV, from the TÜBİTAK UME 10 V PJVS system, with the following experimental setup:

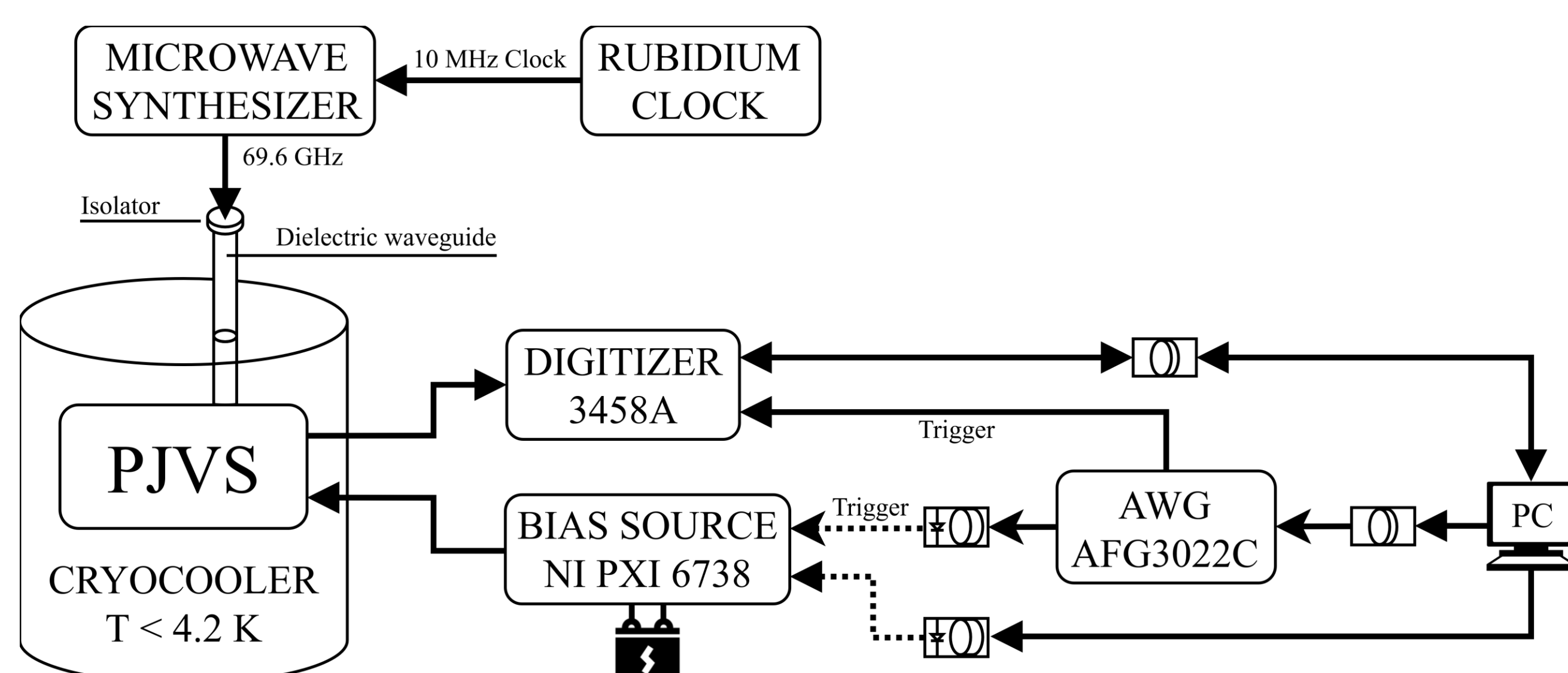


Fig. 1 – Experimental setup for DC voltage measurements with PJVS system

A neural network is trained on the system's measurements, and the model's prediction bias is obtained as the mean of the residuals between reference measurements and predictions, from which we can estimate the model bias uncertainty. The total variance is decomposed using the law of total variance.

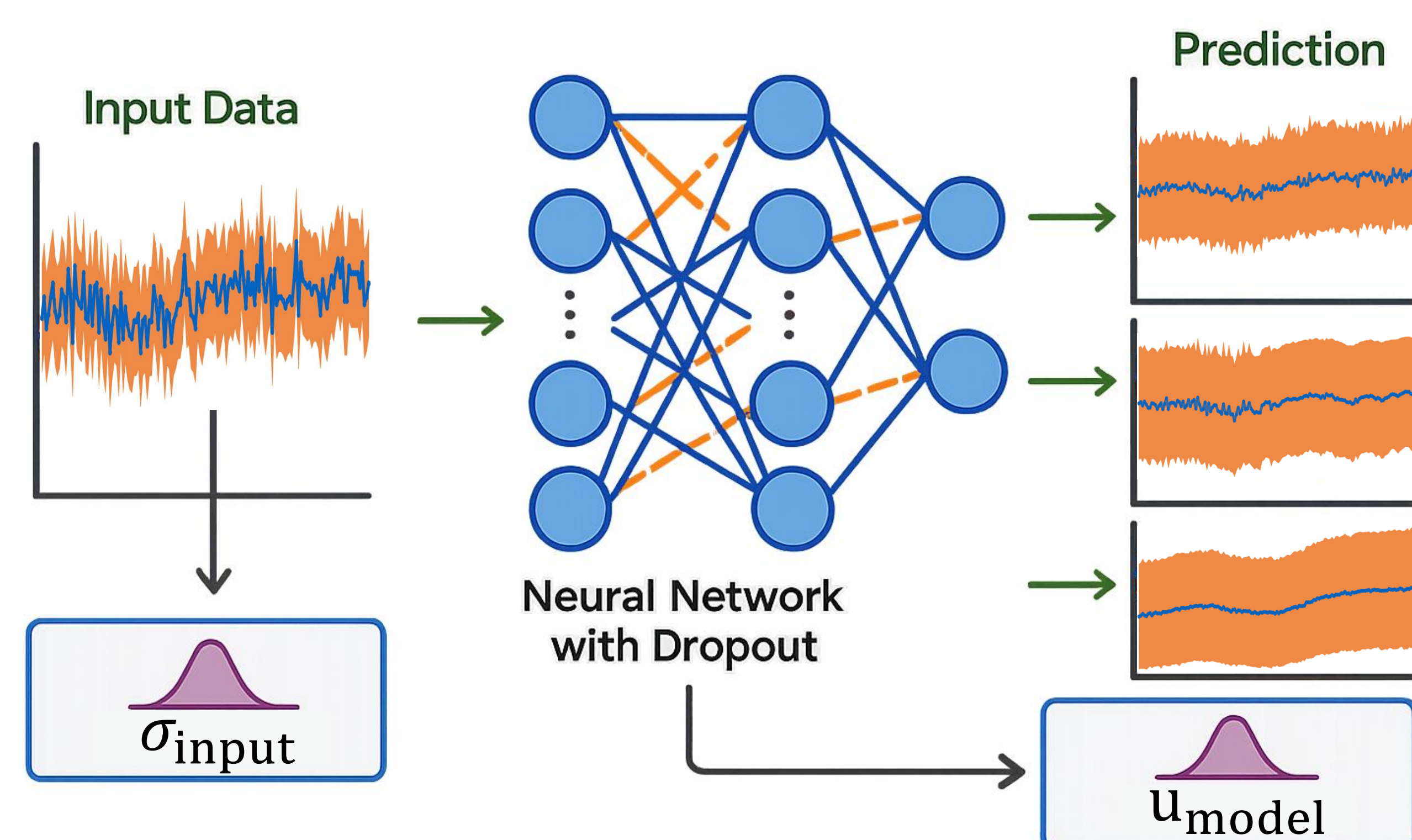


Fig. 2 – Combination of Monte Carlo and Monte Carlo Dropout methods

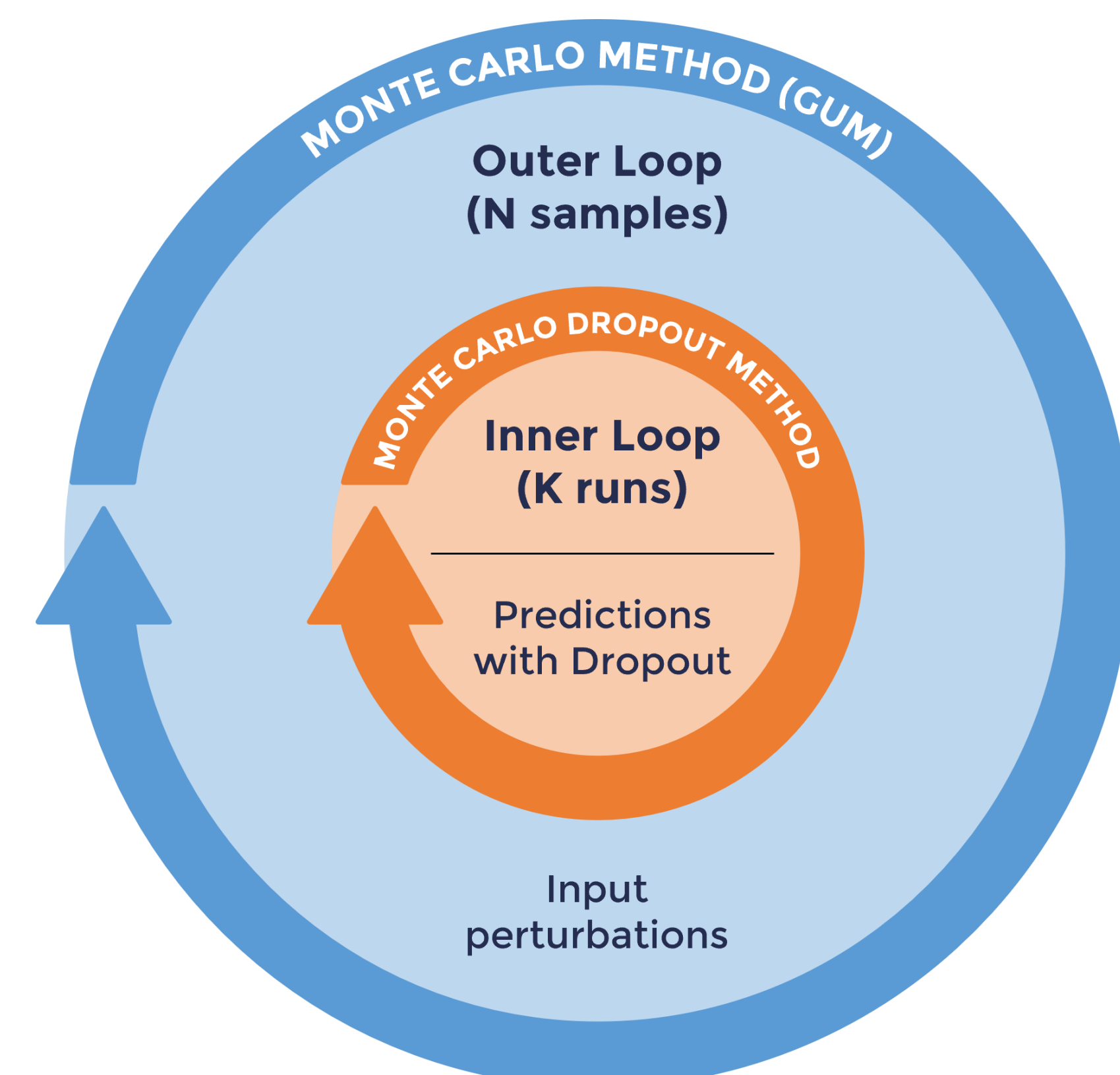


Fig. 3 – Representation of the double loop uncertainty evaluation method

RESULTS

Three architectures were tested as surrogate models of the PJVS digital twin: a multilayer perceptron (MLP), a long short-term memory network (LSTM), and a convolutional neural network (CNN). All models were trained separating in 70% training, 20% validation, and 10% test split in a one-step-ahead sequence-to-sequence setting. The DMCUF used $N = 1000$ Monte Carlo input samples with an uncertainty $\sigma_{\text{input}} = 79$ nV estimated from the measurement series, and $K = 200$ Monte Carlo Dropout realizations (dropout rate = 40%). The coverage factor is $k = 1$.

Table 1 - Comparison of predictive performance (test set)

Model	RMSE (nV)	Bias (nV)	Coverage 95%
MLP	64	-2	96.47%
LSTM	66	-13	90.93%
CNN	69	-17	73.05%

Table 2 - Comparison of uncertainty components (test set)

Model	u_b (nV)	$\langle u_{\text{input}} \rangle$ (nV)	$\langle u_{\text{model}} \rangle$ (nV)	$\langle u_c \rangle$ (nV)
MLP	3	22	67	71
LSTM	3	16	50	53
CNN	3	9	33	35

CONCLUSIONS

DMCUF provides a GUM-compliant uncertainty evaluation method for neural-network-based models, as demonstrated in this PJVS digital twin application. It provides a transparent and reproducible way to quantify uncertainty contributions from both inputs and model structure. Results showed that combined uncertainty is strongly dependent on model architecture and dropout layer configuration. Due to its superior predictive performance, the MLP model proved to be the one that best represents the measured data. The approach presented is easily applicable to other metrological systems that employ data-driven models.

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